

La clausura de Zariski es el conjunto de ceros del ideal de anulaci3n

Lema 1. Sea $S \subset \mathbb{A}_K^n$, entonces $\overline{S} = \mathbb{V}(\mathbb{I}(S))$ donde $\overline{S}$ es la clausura de Zariski de S .

Demostraci3n: Como la intersecci3n de variedades algebraicas afines es una variedad algebraica afn (v3ase la demostraci3n de la [prop-topologia-zariski/??](#)), tenemos que $\overline{S} = \bigcap_{Y \supset S \atop Y \text{ v.a.a.}} Y$ y, por la [prop-variedad-algebraica-afin-ideal/??](#), $\exists J \subset K[x_1, \dots, x_n]$ ideal tal que $\overline{S} = \mathbb{V}(J)$. Entonces, $J \subset \mathbb{I}(S)$ y, al aplicar $\mathbb{V}$, deducimos que $\overline{S} = \mathbb{V}(J) \supset \mathbb{V}(\mathbb{I}(S))$.

Como, por otro lado, $\mathbb{V}(\mathbb{I}(S))$ es una variedad algebraica afn que contiene a S , por definici3n de $\overline{S}$, tenemos que $\overline{S} \subset \mathbb{V}(\mathbb{I}(S))$. Por tanto, $\overline{S} = \mathbb{V}(\mathbb{I}(S))$. ■

Referenciado en

- [teo-clausura-zariski-morfismo-variedades-algebraicas-afines](#)

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